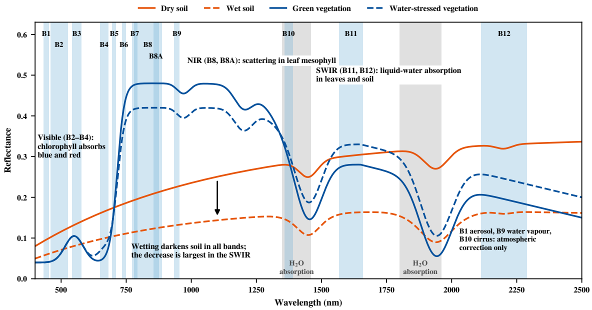
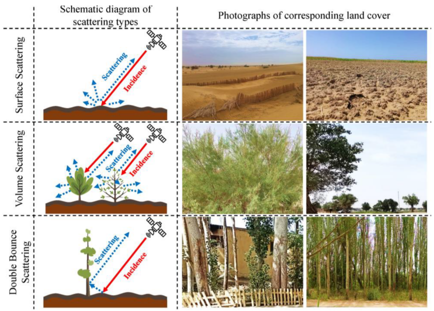
Motivation Different ways of obtaining soil moisture

Source	How SM is obtained	Depth	Main limitation
Microwave (SMAP, SMOS, ASCAT, S1)	invert: σ^0 or $T_B = f(\mathrm{SM}, \mathrm{roughness}, \mathrm{vegetation})$	top ∼5 cm; informs part of the root zone (Feldman et al., 2023)	shallow; assumes vegetation and roughness; coarse (9–40 km) except S1
Optical / thermal (S2, Landsat, MODIS)	empirical relations with surface properties: reflectance and indices; LST–vegetation space (e.g. triangle method, TVDI)	skin / surface	indirect, site-specific calibration; clouds; canopy hides the soil
Land-surface models	$d\theta/dt = P - ET - Q - D$, with parameters p	full profile	forcing errors, soil and vegetation parameters, process representation (Koster et al., 2009; Seneviratne et al., 2010)
Reanalysis (ERA5-Land)	LSM driven by reanalysis meteorology	full profile	model assumptions dominate deep layers with few observational constraints (Li et al., 2021; Miguez-Macho and Fan, 2021)
In situ (ISMN)	direct probe measurement	several depths	point scale; sparse, uneven global coverage
Data-driven (ML / DL)	$\mathrm{SM} = f_\theta(\mathrm{inputs})$, learned from in-situ labels; no ad-hoc process assumptions	any depth with labels	needs enough, representative sites; generalisation?; hard to interpret?

Motivation **Multimodal remote sensing**



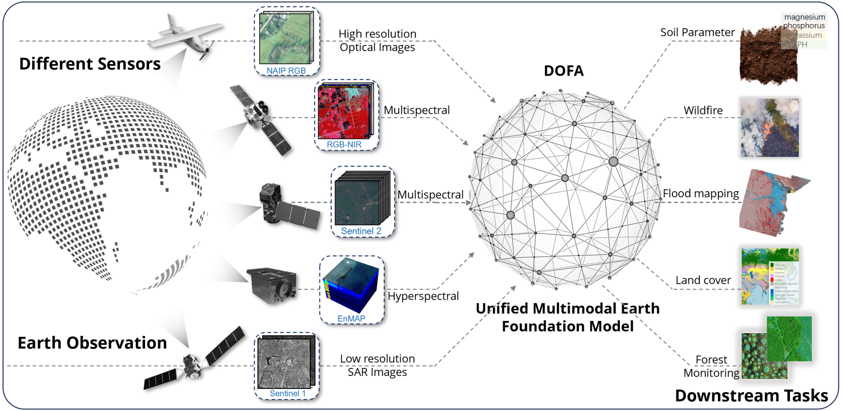
Optical: dry vs wet soil; green vs water-stressed vegetation


Schematic diagram of scattering types
Photographs of corresponding land cover
Surface Scattering
Volume Scattering
Double Bounce Scattering
Scattering
Incidence
Scattering
Incidence
Scattering
Incidence

SAR scattering types (Nurmemet et al., 2025)

Soil and vegetation water change both optical reflectance and SAR backscatter

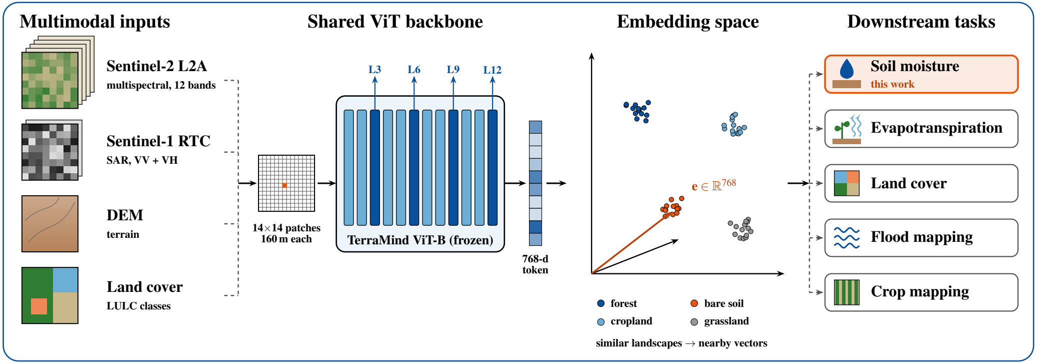
Motivation Geospatial foundation models (GFMs)


Different Sensors
Earth Observation
NAIP RGB
High resolution
Optical Images
RGB-NIR
Multispectral
Sentinel 2
Multispectral
EnMAP
Hyperspectral
Sentinel 1
Low resolution
SAR Images
DOFA
Unified Multimodal Earth
Foundation Model
Soil Parameter
magnesium
phosphorus
assium
PH
Wildfire
Flood mapping
Land cover
Forest
Monitoring
Downstream Tasks

DOFA: one model for many sensors and many downstream tasks (Xiong et al., 2024, arXiv:2403.15356)

Once pre-trained on multimodal, unlabelled data, it works for many downstream tasks

Motivation TerraMind embeddings for soil moisture



Multimodal inputs

Sentinel-2 L2A
multispectral, 12 bands

Sentinel-1 RTC
SAR, VV + VH

DEM
terrain

Land cover
LULC classes

14×14 patches
160 m each

Shared ViT backbone

L3 L6 L9 L12

TerraMind ViT-B (frozen)

768-d
token

Embedding space

$\mathbf{e} \in \mathbb{R}^{768}$

● forest
● bare soil
● cropland
● grassland

similar landscapes → nearby vectors

Downstream tasks

Soil moisture
this work

Evapotranspiration

Land cover

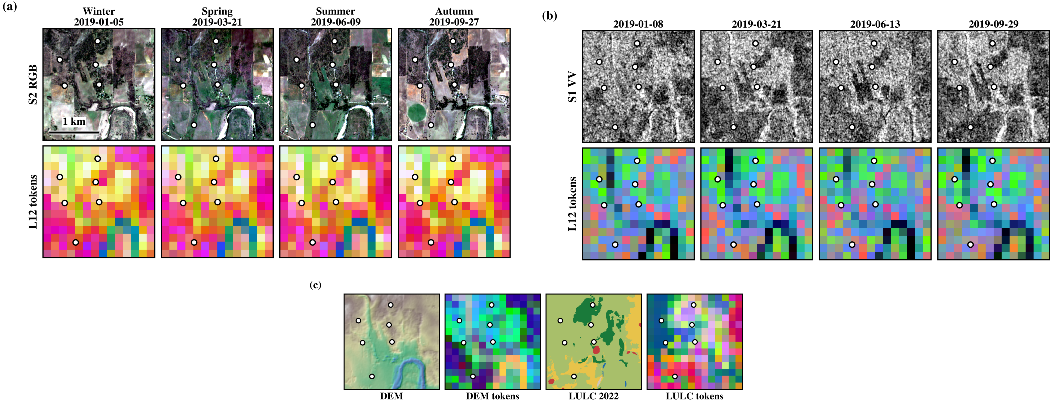
Flood mapping

Crop mapping

S2, S1, DEM and LULC → frozen TerraMind ViT-B → one 768-d embedding per 160 m patch → downstream tasks

The same embedding can serve many tasks; in this case, soil moisture

Motivation Example: TerraMind embeddings at one site (TxSON, 2019)


(a)
Winter
2019-01-05
Spring
2019-03-21
Summer
2019-06-09
Autumn
2019-09-27
S2 RGB
1 km
L12 tokens
(b)
2019-01-08
2019-03-21
2019-06-13
2019-09-29
S1 VV
L12 tokens
(c)
DEM
DEM tokens
LULC 2022
LULC tokens

(a) S2 in four seasons, (b) S1 VV, (c) DEM and LULC, each with its TerraMind L12 tokens

Each 160 m patch → one 768-d vector; colours show its first 3 principal components